

Binary to decimal conversions

$$\textcircled{1} \begin{array}{r} 1011 \\ \underline{2} \end{array} \Rightarrow \begin{array}{r} 8 \ 4 \ 2 \ 1 \\ 1 \ 0 \ 1 \ 1 \end{array} \Rightarrow 8 + 2 + 1 = 11_{10}$$

$$\textcircled{2} \begin{array}{r} 11010 \\ \underline{2} \end{array} \Rightarrow \begin{array}{r} 16 \ 8 \ 4 \ 2 \ 1 \\ 1 \ 1 \ 0 \ 1 \ 0 \end{array} \Rightarrow 16 + 8 + 2 = 26_{10}$$

$$\textcircled{3} \begin{array}{r} 100111 \\ \underline{2} \end{array} \Rightarrow \begin{array}{r} 32 \ 16 \ 8 \ 4 \ 2 \ 1 \\ 1 \ 0 \ 0 \ 1 \ 1 \ 1 \end{array} \Rightarrow 32 + 4 + 2 + 1 = 39_{10}$$

$$\textcircled{4} \begin{array}{r} 111000 \\ \underline{2} \end{array} \Rightarrow \begin{array}{r} 32 \ 16 \ 8 \ 4 \ 2 \ 1 \\ 1 \ 1 \ 1 \ 0 \ 0 \ 0 \end{array} \Rightarrow 32 + 16 + 8 = 56_{10}$$

$$\textcircled{5} \begin{array}{r} 1010101 \\ \underline{2} \end{array} \Rightarrow \begin{array}{r} 64 \ 32 \ 16 \ 8 \ 4 \ 2 \ 1 \\ 1 \ 0 \ 1 \ 0 \ 1 \ 0 \ 1 \end{array} \Rightarrow 64 + 16 + 4 + 1 = 85_{10}$$

$$\textcircled{6} \begin{array}{r} 11001100 \\ \underline{2} \end{array} \Rightarrow \begin{array}{r} 128 \ 64 \ 32 \ 16 \ 8 \ 4 \ 2 \ 1 \\ 1 \ 1 \ 0 \ 0 \ 1 \ 1 \ 0 \ 0 \end{array} \Rightarrow 128 + 64 + 8 + 4 = 204_{10}$$

$$\textcircled{7} \begin{array}{r} 10000001 \\ \underline{2} \end{array} \Rightarrow \begin{array}{r} 128 \ 64 \ 32 \ 16 \ 8 \ 4 \ 2 \ 1 \\ 1 \ 0 \ 0 \ 0 \ 0 \ 0 \ 0 \ 1 \end{array} \Rightarrow 128 + 1 = 129_{10}$$

$$\textcircled{8} \begin{array}{r} 10110110 \\ \underline{2} \end{array} \Rightarrow \begin{array}{r} 128 \ 64 \ 32 \ 16 \ 8 \ 4 \ 2 \ 1 \\ 1 \ 0 \ 1 \ 1 \ 0 \ 1 \ 1 \ 0 \end{array} \Rightarrow 128 + 16 + 8 + 4 = 156_{10}$$

9) $1111111 \rightarrow 128 + 64 + 32 + 16 + 8 + 4 + 2 + 1$
 $128 \ 64 \ 32 \ 16 \ 8 \ 4 \ 2 \ 1$
 $= 2^8 - 1 = 255_{10}$

10) $110100111 \rightarrow 256 + 128 + 32 + 4 + 2 + 1 = 423_{10}$
 $256 \ 128 \ 64 \ 32 \ 16 \ 8 \ 4 \ 2 \ 1$

Decimal to Binary conversions

1) $14 \rightarrow 8 \ 4 \ 2 \ 1 \Rightarrow 1110_2$
 $< 15 \rightarrow 2^4 - 1$
 $1 \ 1 \ 1 \ 0$

2) $27 \rightarrow 16 \ 8 \ 4 \ 2 \ 1 \rightarrow 11011_2$
 $< 31 \rightarrow 2^5 - 1$
 $1 \ 1 \ 0 \ 1 \ 1$

3) $99 \rightarrow 64 \ 32 \ 16 \ 8 \ 4 \ 2 \ 1 \rightarrow 1100011_2$
 $< 127 \rightarrow 2^7 - 1$
 $1 \ 1 \ 0 \ 0 \ 0 \ 1 \ 1$

4) $312 \rightarrow 256 \ 128 \ 64 \ 32 \ 16 \ 8 \ 4 \ 2 \ 1 \rightarrow 100101100_2$
 $< 511 \rightarrow 2^9 - 1$
 $1 \ 0 \ 0 \ 1 \ 0 \ 1 \ 1 \ 0 \ 0$

5) $1024 \rightarrow 1024 \ 512 \ 256 \ 128 \ 64 \ 32 \ 16 \ 8 \ 4 \ 2 \ 1 \rightarrow 1000000000_2$
 $< 2047 \rightarrow 2^{10} - 1$
 $1 \ 0 \ 0 \ 0 \ 0 \ 0 \ 0 \ 0 \ 0 \ 0 \ 0$
 $1111111111 + 1$
 $= 10000000000_2$

Binary to Hex Conversion

① $\underline{1101}_2 \rightarrow 2_{16}$

② $\begin{matrix} 1101 \\ 8421 \end{matrix} \rightarrow 13 \rightarrow D_{16}$

③ $\underline{101011}_2$

$\begin{matrix} 0010 & 1011 \\ \hline \downarrow & \downarrow \\ 2 & 11 \end{matrix} \rightarrow 2B_{16}$

④ $\begin{matrix} 1001 & 1101 \\ \hline 2421 & 8421 \end{matrix} \rightarrow 9B_{16}$

⑤ $\begin{matrix} 101010 & 101010 \\ \hline 10 & 10 \end{matrix} \rightarrow AAA_{16}$

⑥ $\begin{matrix} 1111010 & 11001110 \\ \hline 15 & 12 & 14 \end{matrix} \Rightarrow FACE_{16}$

Hex to Binary conversion

$$\textcircled{1} \quad 4A_{16} \rightarrow 01001010_2$$

0100 1010

$$\textcircled{2} \quad 1B7 \rightarrow 000110110111_2$$

0001 1011 0111

$$\textcircled{3} \quad 8DF \rightarrow 100011011111_2$$

1000 1101 1111

$$\textcircled{4} \quad 2E4B \rightarrow 0010111001001011_2$$

0010 1110 0100 1011

Octal - Decimal conversion

$$\textcircled{1} \quad 17 \Rightarrow 001111_2 \Rightarrow 15_{10}$$

421 421
001 111

$$\textcircled{2} \quad 3526_8 \Rightarrow ? \quad 3 \times 8^3 + 5 \times 8^2 + 2 \times 8^1 + 6 \times 8^0$$
$$= 1878_{10}$$

$$\textcircled{3} \quad 123_8 \Rightarrow ? \quad 1 \times 8^2 + 2 \times 8^1 + 3 \times 8^0$$
$$64 + 16 + 3 \Rightarrow 83_{10}$$

Decimal $\rightarrow$ Octal conversion

① $24_{10} \rightarrow ?_8$
 $= 30_8$

$$8 \overline{) 24} (3$$

$$\underline{24}$$

$$\underline{0}$$

$$8 \overline{) 30} (3$$

$$\underline{24}$$

$$\underline{6}$$

② $79_{10} \rightarrow ?_8$
 $= 117_8$

$$8 \overline{) 79} (9$$

$$\underline{72}$$

$$\underline{7}$$

$$8 \overline{) 91} (11$$

$$\underline{8}$$

$$\underline{1}$$

$$8 \overline{) 101} (12$$

$$\underline{96}$$

$$\underline{5}$$

③ $320_{10} \rightarrow ?_8$
 $= 500_8$

$$8 \overline{) 320} (40$$

$$\underline{320}$$

$$\underline{0}$$

$$8 \overline{) 4015} (501$$

$$\underline{4000}$$

$$\underline{15}$$

$$8 \overline{) 5000} (625$$

$$\underline{4800}$$

$$\underline{200}$$

Signed Binary (2's complement) $\rightarrow$ Decimal conversions

① 0110 (4 bit)

$-8421 \rightarrow$

MSB = 0 $\Rightarrow +ve$

0110

$-8421 \Rightarrow +6$

② 1010

-8421

$\rightarrow$

MSB = 1 $\Rightarrow -ve$

1010

$= -8 + 2 = -6$

-8421